

Ashutosh Soni

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Areas of interest: Transfer Learning, Deep Reinforcement Learning, Large Language Models, Generative AI.

EDUCATION

- **The Ohio State University: GPA 3.63/4.00** Columbus, OH
Ph.D. in Department of Electrical and Computer Engineering (ECE) Aug 2022 – Present
- **Indian Institute of Technology (IIT), Bhilai: GPA 9.49/10.00** Bhilai, India
Bachelor of Technology (Honours) in Electrical Engineering (Senate Award for Highest GPA) Aug 2018 – May 2022

RESEARCH EXPERIENCE

- **Graduate Research Assistant, The Ohio State University** Columbus, OH
Supervisors - Prof. Ness Shroff and Prof. Atilla Eryilmaz Aug 2022 – Present
 - Developed a novel task-similarity metric (BeST) to efficiently identify the most transferable pre-trained source models for a given target task in transfer learning without additional training. We use an innovative quantization-based optimization procedure to calculate the metric, only using the source models and target data.
 - Worked on the analytical formulation of optimal train and corresponding test transfer accuracy. Our metric only requires one pass of the target data through the pre-trained source model.
 - The metric takes significantly less time to find the most transferable source model as compared to classical transfer learning training, with an improvement factor ranging from $\times 5$ to $\times 50$. Datasets used - MNIST, CIFAR10, Imagenette. Even for 100 samples, our metric can predict the best source with $> 80\%$ accuracy. ([ArXiv preprint](#)).
- **Undergraduate Researcher for B.Tech. (Honours) Thesis, IIT Bhilai** Bhilai, India
Supervisors - Prof. Arzad Alam Kherani and Prof. Sreejith TV, IIT Bhilai Aug 2020 – May 2022
 - **Reconfigurable Intelligent Surface (RIS):** Studied the performance of RIS-assisted networks for noise-limited cases. Characterized SNR and studied its variation with the density of RIS and mobility of users. Learned to use [SimRIS](#) channel simulator in MATLAB. Were able to show improvements of the order of 80-400 % when using RIS. We also show the existence of an optimal radius of RIS deployment that can maximize SNR gains ([Paper link](#)).
 - **Joint Transmission in Wi-Fi:** As part of **research fellowship** from MOJO Network Pvt Ltd., studied multiple access point transmission for Wi-Fi 7 (IEEE 802.11be) and developed a scheduling algorithm. Used queuing theory to analyze queue lengths and jitter, and theorized conditions for stability.
- **Research Internship at Indian Institute of Science (IISc), Bangalore** Bangalore, India
Supervisor - Prof. Vinod Sharma, EECS Dept., IISc Bangalore May 2020 – July 2020
 - Worked to implement and study the performance of the Deep Deterministic Policy Gradient (DDPG) algorithm to choose an optimal transmission power for a multicast system.
 - Learned to write complex reinforcement learning environments and code algorithms like DQN and DDPG.
 - Our results suggest that DDPG works better than DQN for optimal power control in terms of better sojourn times.
- **Research Internship at Sooktha Consulting Pvt. Ltd.** Bangalore, India
Supervisor - Balaji Rangaswamy (CEO), Sooktha Consulting Pvt. Ltd. May 2021 – July 2021
 - Studied and implemented prior algorithms for processing the uplink Sounding Reference Signal (SRS) in 5G NR technology to determine the beamforming weights for MU-MIMO transmissions in the downlink.
 - Learned to use 5G NR toolbox in MATLAB to implement and compare different algorithms.
 - This work formed the basis for the implementation of a ma-MIMO/MU-MIMO scheduler in the company's 5G NR base station (gNB) software solution.

SKILLS

- **Programming Languages:** Python, C, C++, MATLAB
- **ML Frameworks:** Tensorflow, Keras, Scipy, NumPy, Pandas, Scikit-learn
- **Miscellaneous:** Git, $\text{\LaTeX}$

PUBLICATIONS

- Kausheek, Akella SK, **Ashutosh Soni**, T. V. Sreejith, and Arzad A. Kherani. "Performance of RIS assisted networks in noise limited and interference limited scenarios." In 2022 14th International Conference on COMMunication Systems & NETworkS (COMSNETS), pp. 565-573. IEEE, 2022 ([IEEEExplore](#)).
- **Ashutosh Soni**, Peizhong Ju, Atilla Eryilmaz, and Ness B. Shroff. "BeST—A Novel Source Selection Metric for Transfer Learning." arXiv preprint arXiv:2501.10933 (2025), ([ArXiv](#)).